

SleepABP: Noninvasive Ambulatory Blood Pressure Monitoring Based on Ballistocardiogram in Sleep State

Zhi, Tian

Northwestern Polytechnical University, Xi'an China
zhi.tian@mail.nwpu.edu.cn

Xiaoguang, Yu

Northwestern Polytechnical University, Xi'an China
yxiaoxiao@mail.nwpu.edu.cn

Hongbo, Ni

Northwestern Polytechnical University, Xi'an China
nihb@nwpu.edu.cn

Alex Q., Chen

Singapore Institute of Technology, Singapore
Alex.Q.Chen@singaporetech.edu.sg

ABSTRACT

Blood pressure (BP) is one of the vital sign signals used to evaluate the health status of the human body, and continuous ambulatory blood pressure (ABP) monitoring can accurately and comprehensively reflect the physiological status of the cardiovascular system. ABP is mainly measured using ambulatory sphygmomanometer, by regularly collecting the pressure changes in blood vessels. However, this method is usually intrusive, inconvenient, and expensive. In this paper, we develop SleepABP, a novel natural, continuous and accurate ABP monitoring system for daily sleep scenarios. SleepABP is based on the Ballistocardiogram (BCG) signal, which relies on measuring the tiny fluctuations (impulses) in the body caused by the continuous beating of the heart. Specifically, the proposed BCG sensing system monitors the pulses from the user's head applying the piezoelectric ceramic sensor. The BCG signals are recorded simultaneously, and the BCG cycle restoration algorithm is used to extract the continuous and complete BCG cycles for constructing the multi-dimensional features of the blood pressure prediction. SleepABP was evaluated on 16 participants under various sleeping conditions, positions, and personal internal factors. The RMSE results of diastolic blood pressure 3.22 mmHg and systolic blood pressure 3.5 mmHg demonstrates the system's effectiveness and feasibility in a daily sleeping environment.

CCS CONCEPTS

• : **Human-centered computing** → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Ballistocardiography (BCG), Nonintrusive Ambulatory Blood Pressure, Multidimensional Features, Physiological Perception

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1 INTRODUCTION

Sleep ambulatory blood pressure refers to the blood pressure measured at regular intervals during sleep [1]. People in modern society, facing problems such as unhealthy lifestyle, irregular diet and high work pressure, have a weak health consciousness, which increases the risk of cardiovascular disease. There is an urgent need for a non-contact, non-intrusive, continuous and stable sleep ambulatory blood pressure monitoring method. At present, ambulatory blood pressure monitoring relies on cuff and finger cuffs for measurements or blood pressure measurement using photoplethysmography (PPG) signals. Nevertheless, these methods interfere with the user's rest and are inconvenient to use. Hence, it is impractical for the user to continuously and stably monitor the ambulatory blood pressure during sleep for a prolonged period of time.

This study proposes a continuous ambulatory blood pressure monitoring system called SleepABP. The SleepABP consists of a piezo sensor and an STM32 chip (Figure 1). When the user is sleeping, the piezoelectric ceramic sensor of SleepABP collects the user's heartbeat signal and obtains the head pulse signal [2], thereby extracting the crossbow electrocardiogram (BCG) signal. Compared with the impact of using photoplethysmography (PPG) on the user, the monitoring process of SleepABP is completely non-interventional for the user and does not need to be directly attached to the human body. Physiological parameters of the user may be monitored without user perception, and. At the same time, the interference of the cuff on blood vessels and surrounding tissues is completely avoided. The blood pressure estimation (prediction) model in SleepABP is based on the principle of BCG generation, the correlation mechanism between BCG and blood pressure, and hemodynamics. For example, features such as transit time between point I and point J in BCG have been well verified to be associated with BP [3, 4] Therefore, relying on BCG and cardiovascular multidimensional features, blood pressure can be estimated continuously.

The main contributions of this work are the innovation of application scenarios, using BCG sensing technology for non-intervention continuous blood pressure monitoring during sleep. The contributions include 1) Using an array piezoelectric ceramic sensor to collect physiological signal during sleep; 2) A BCG recovery recognition detection algorithm is proposed to obtain continuous and

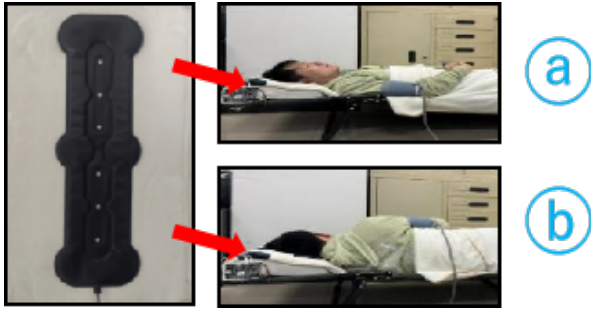


Figure 1: ABP monitoring using SleepABP during normal sleep (a. supine sleeping position; b. side sleeping position)

complete BCG cycle signals, which improves the robustness of BCG acquisition under different sleeping positions; 3) The extraction of multi-dimensional features for achieving the ABP estimation.

2 RELATED WORK

2.1 BCG Signal Acquisition and Application

BCG is a zero-load measurement technique that detects weak pressure changes originating from body vibrations that are caused by heart beats. When the heart contracts/diastoles, the sensor can sense the force that the blood hits the blood vessel to produce the force opposite to the blood flow, and this change reflects the volume and speed change produced by each beat of the heart. Compared with the cuff-type ambulatory blood pressure method, BCG measurement of blood pressure does not affect the user's rest, and can continuously monitor sleep blood pressure. This makes the relationship between BCG transmission principle and BP change widely concerned [3–6].

BCG and blood pressure waveforms are inextricably linked due to the principle of pulse transmission (Figure 2). Blood pressure is measured at various locations to calculate the BP gradients of the ascending aorta and descending aorta of these waveforms to reconstruct the BCG wave. This process establishes the relationship between BCG and BP, and provides a theoretical basis for BCG to measure blood pressure. However, in the scenario where only BCG is used to measure blood pressure, there are partial applications in standing and sitting postures [15, 16], while in lying postures it is limited to the measurement of respiration and heart rate [18] while ambulatory blood pressure monitoring [8, 17] is rarely used.

2.2 Current Status of Non-contact Blood Pressure Monitoring

Non-contact blood pressure monitoring is of great significance to modern people. It has been realized in many scenarios. However, due to the limitations of theory and equipment, there is no in-depth research on non-contact blood pressure monitoring during sleep. Most studies adopt BCG, ECG, and other signals for blood pressure monitoring during sleep [7] and other applications [8, 9]. Comparing these studies with non-contact blood pressure monitoring in the office environment [12], collecting iPPG by camera is not suitable for sleeping scenario use cases. The desire to reduce physical constraints has inspired the research on BCG. Sensors are placed under

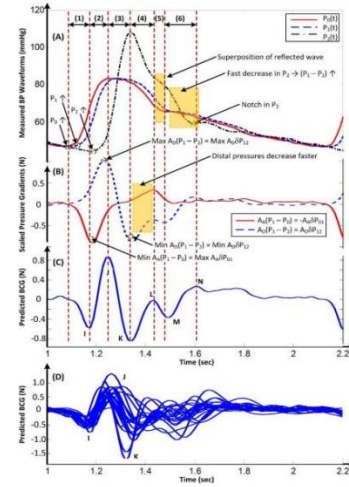


Figure 2: Blood pressure wave & Ballistocardiogram signal (P0/P1/P2 are the blood pressure waves at the aortic inlet, aortic arch top and aortic outlet respectively) [5]

the mattress to collect the BCG signals from the user's back when lying still to estimate the blood pressure, and verify the feasibility of the BCG blood pressure measurement [3]. In SleepABP, besides validated IJ time, JK amplitude [3, 10], we extract the novel multi-dimensional features based on BCG generated during sleep, and then estimate ambulatory blood pressure.

3 IMPLEMENTATION

SleepABP implements a BCG measurement system using common and affordable electronic component. Meanwhile, we evaluate the feasibility of BCG acquisition in various sleeping positions, extracted the complete BCG cycles after preprocessing, extracted multi-dimensional features, and established an estimation model. An overview of the SleepABP framework is shown in Figure 3.

3.1 Hardware Deployment

SleepABP implants a piezoelectric sensor (sampling rate, 100Hz) into the pillow to obtain BCG signals during sleep (Figure 1), which is connected to the chip via USB and does not require direct power supply to avoid motion interference, battery capacity limitations, and safety issues for most wearable devices. Digital signal processing is implemented on an STM32 microcontroller, and the collected data is transmitted to a cloud server at certain intervals via WiFi or 4G modules to obtain real-time digital signals.

3.2 Extraction of BCG

The head BCG signals collected from piezoelectric sensor will exist a baseline drift due to environmental and device noise. To eliminate the diverse noises in the signal, a low-pass filter is used, and then an original signal containing respiration, body motion and BCG signals is obtained. The basic principle of the piezoelectric sensor to obtain BCG is that with each heartbeat, the body will produce a weak force, causing the voltage on the piezoelectric sensor to change. These changes indicate the cycle of cardiovascular time, which can

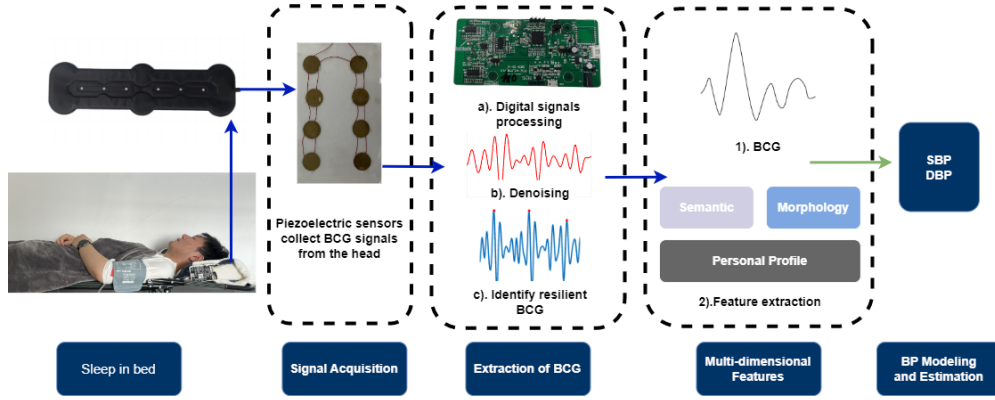


Figure 3: SleepABP Framework

be passed through recording arterial pressure changes and convert them into digital waves, and recover BCG waveforms from digital waves for monitoring. Hence, we should obtain pressure changes from the carotid artery to extract BCG. The processing procedure includes two main steps: denoise and reconstruct BCG waveform; identify and restore BCG cycle.

Denoise & reconstruct BCG waveform: The purpose of this process is to obtain the BCG information produced by each heart-beat of the subject, which is called the preliminary BCG information (PBCG). By decomposing the original waveform to remove the interference of irrelevant areas such as breathing and body movement, the quality of the BCG waveform is improved, and the remaining signal is reconstructed to obtain PBCG. In SleepABP, wavelet decomposition is used to remove various interference factors, and then PBCG is obtained by wavelet reconstruction.

Identify and restore BCG cycle: The obtained PBCG still contains non-breathing and body movement-induced interference factors that affect the quality of BCG. This step distinguishes the cycle caused by heartbeat and non-heartbeat and obtains the Refined BCG signal (RBCG). A standard method to extract the BCG cycle is to segment the BCG signal using the R peak of ECG as a synchronous signal and then detect the J peak and BCG cycle Blind Segmentation Algorithm (BBSA) [11]. However, BBSA relies on empirical values and multi-instance learning, and the above methods all need auxiliary means to identify BCG cycles. BCG is not a completely regular signal, and adjacent BCG signals that are not entirely consistent in shape often have randomness and different details [11]. Therefore, not all cycles in PBCG are RBCG cycles. SleepABP proposes a Cycle Recovery Algorithm (CRA) based on BCG cycle characteristics and heart rate to extract RBCG cycles. According to the invariable characteristics of BCG (the number of peaks and valleys, etc.) and heart rate, the PBCG is identified and restored, and the RBCG signal is restored. Applying the CRA can remove interference from PBCG information and accurately identify and restore the RBCG signal (Figure 4).

3.3 Multidimensional Features

According to the biomedical knowledge [12], the characteristics of BCG are extracted from four dimensions:

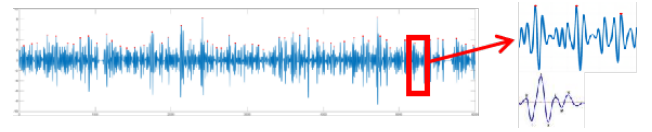


Figure 4: Final BCG cycle identification (a. identified BCG cycle; b. standard BCG cycle)

First dimension: Based on the principle of BCG generation (physiological state represented by waveform changes, with definite clinical significance): including instantaneous heart rate (HR), heart rate variability feature set, Ballistocardiogram Intensity Ratio (BIR: the amplitude interval between the pulse wave systole and diastole reflects the relationship between the elasticity of blood vessels and blood pressure);

Second dimension: Based on the correlation mechanism between BCG and blood pressure: IJ pulse transit time, IJ pulse amplitude difference;

Third dimension: Morphological features based on hemodynamics: BCG morphology shows that the change of blood flow under the influence of blood pressure leads to the change of waveform, including the characteristics of different positions of BCG such as IJ pulse slope, JK pulse transmission time, and JK pulse amplitude difference and systolic area;

Fourth dimension: Personal data characteristics: gender, age, BMI, etc. It is worth mentioning that the BCG in SleepABP contains a lot of detailed information, the level of systolic blood pressure reflects the amount of cardiac output per stroke and the elasticity of the blood vessel wall [13, 14], and the shape and size of the area formed between IJK are related to systolic pressure. Therefore, in SleepABP, systolic period area (as shown in Figure 5) is adopted as an important feature for predicting BP.

4 EXPERIMENT AND EVALUATION

4.1 Experimental Setup

Sixteen volunteers were recruited for the test experiment, and their demographics are shown in Table 1. The experimental data was collected twice daily: once at night and the other during daytime

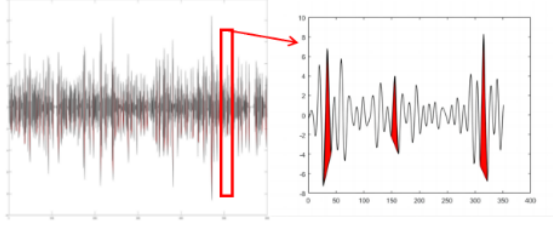


Figure 5: Multidimensional characteristic systolic area

naps. Volunteers are asked simulated two sleeping positions: lying on the back and lying on their side (either left or right side respectively). Both sleeping positions are intended to demonstrate the robustness of SleepABP monitoring blood pressure. During the experiment, volunteers can perform normal sleep activities such as turning over. Labeled blood pressure was measured every minute using a standard cuff blood pressure device (Omron Hem-7124). We apply the dataset collected by Carlson Charles [3] as training set and validation set, using head BCG in the dataset and ambulatory blood pressure for calibration. After calibration, the data from the volunteers were used for testing, and their blood pressure during sleep was estimated according to the blood pressure model, and compared with the labelled blood pressure.

4.2 Evaluation Results

BCG cycle reconstitution & multi-dimensional features: Compared with the existing CRA used by BCG in two representative sleeping positions, BBSA determines the BCG cycle through information such as extreme points and time intervals, which is used to accurately describe the heart rate [11]. CRA not only uses extreme points to find the BCG cycle, but also reconstructs the BCG waveform by describing the detailed information of BCG, so as to extract multi-dimensional features, such as systolic period area (Figure 5), etc., to evaluate ambulatory blood pressure.

Blood pressure estimation: We have a total of $m=40$ (40 volunteers, about 4.65 hours in total) BCG and blood pressure in the training set, and $n = 64$ (16 volunteers $\times$ 2 sleeping positions $\times$ 2 times) blood pressure and BCG in the test set for analysis. After verifying the different regression models of BP estimation, we choose the random forest model, which has the highest estimation

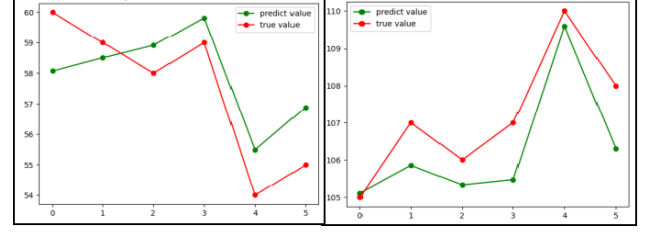


Figure 7: Test set blood pressure evaluation map

accuracy, and randomly select some people in the data set as the verification set, as shown in Figure 6; in the final root mean square error (RMSE) results of all participants we achieving a DBP of 3.22 mmHg and a SBP of 3.50 mmHg (Fig. 7 is the blood pressure prediction map of a volunteer), and the individual breakdown is shown in Table 1.

5 CONCLUSION

In this study, we proposed SleepABP, a method for non-intervention, continuous ambulatory blood pressure monitoring in the sleep environment of existing equipment, and demonstrated the feasibility of BCG technology for non-intervention, continuous ambulatory blood pressure monitoring in various sleeping positions. The experimental results show that the blood pressure measured by SleepABP is highly consistent with the home medical sphygmomanometer. SleepABP can realize continuous ambulatory blood pressure monitoring during sleep in an ordinary environment to meet people's needs for daily blood pressure monitoring, daily management of cardiovascular diseases and risk warning, and has important application value for self-care health management.

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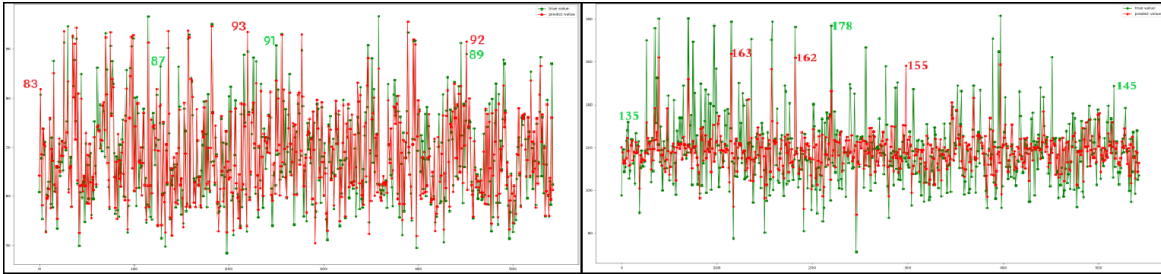


Figure 6: Blood pressure evaluation map of the verification set (the left is the systolic blood pressure, the right is the diastolic blood pressure, the green is the real blood pressure, and the red is the predicted blood pressure)

Table 1: Participant Statistics and Results

ID	Age	Sex	SupineBP[mmHg]	Side-lyingBP[mmHg]	RMSEBP[mmHg]	RMSEBP[mmHg]
1	21	F	57-107	42-90	3.202	2.025
2	22	M	57-119	42-91	4.347	4.358
3	22	M	73-134	65-119	4.081	3.836
4	23	M	53-116	50-97	3.162	4.719
5	23	M	53-114	48-98	4.090	4.616
6	24	M	58-110	58-114	2.749	2.448
7	24	M	54-104	48-85	2.062	2.166
8	24	M	56-106	54-88	1.949	2.314
9	25	F	57-116	48-100	1.866	3.628
10	25	M	44-116	43-89	3.864	4.309
11	25	M	52-119	43-91	4.486	4.958
12	26	F	58-111	41-88	0.953	1.983
13	48	F	57-116	55-100	3.776	3.437
14	49	M	44-116	65-120	4.864	4.257
15	55	F	52-119	60-111	3.157	4.348
16	56	M	58-111	61-117	2.975	2.598
μ	31				3.22	3.5
σ	13				1.06	1.04

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